Ravi Prakash Meghwanshi

Ph.D. Scholar
Department of Electrical Engineering
Indian Institute of Technology (IIT) Jodhpur, Rajasthan, India

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Research Interests

- Human-Computer Interaction
- Human Perception and Psychophysics
- Haptic Interaction
- Interactive and Multimodal Systems
- Perceptual Modeling
- Human-Centered System Design
- User Studies
- Human-AI Interaction

Education

- **Ph.D. | Electrical Engineering (CGPA: 8.54)** *Dec. 2021–Dec. 2026 (expected)*
  Indian Institute of Technology (IIT) Jodhpur, Rajasthan, India
  Thesis: Study of the Factors Related to Force Perception for Telepresence and Teleaction Applications
- **M.Tech. | Control & Instrumentation, Electrical Engineering (CGPA: 8.90)** *Jul. 2017–Jul. 2019*
  Motilal Nehru National Institute of Technology (NIT) Allahabad, Prayagraj, Uttar Pradesh, India
  Thesis: Event-Triggered Output-Feedback Control with Input-Saturation
- **B.E. | Electrical Engineering (Percentage.: 65.80%)** *Jul. 2012–Jul. 2016*
  M.B.M. Engineering College, Jodhpur, India

International Publications

Journal Articles

[J4] Datta K.*, **Meghwanshi R. P.***, Harish R., & Bhardwaj, A. (2026). Bimodal Material Roughness Perception: Haptic-Sound and Visual-Sound. **Perception** (*Under review*).

[J3] **Meghwanshi, R. P.**, Bhardwaj, A., & Kumar, H. (2026). Perceived Intensity Model of Kinesthetic Force in Two-Dimensional Plane. **IEEE Transactions on Human-Machine Systems** (*Under review*).

[J2] **Meghwanshi, R. P.**, Bhardwaj, A., & Kumar, H. (2026). A Study of Deadzone Structure for Multi-Dimensional Haptic Force Perception. **ACM Transactions on Applied Perception**, 23(2), 1–18. doi: 10.1145/3779298

[J1] **Meghwanshi, R. P.**, Bhardwaj, A., & Kumar, H. (2025). Effect of force-rate on continuous kinesthetic force discrimination. **Experimental Brain Research**, 243(5), 118. doi: 10.1007/s00221-025-07080-1

Conference Proceedings

[C3] Srivastava P., **Meghwanshi R. P.**, & Bhardwaj A. (2026). Just Noticeable Difference of SOA for Inter-Finger Apparent Tactile Motion. **2027 CHI Conference on Human Factors in Computing Systems** (*Under review*).

[C2] **Meghwanshi, R. P.**, Bhardwaj, A., & Kumar, H. (2026, August). Perceptual Threshold of Round-Trip Delay in TPTA Systems during Viscoelastic Object Interaction. In Proceedings of the **2026 ACM Symposium on Applied Perception (SAP)** (pp. 1-5). doi: 10.1145/3821409.3821453

[C1] **Meghwanshi, R. P.**, Bhardawj, A., & Kumar, H. (2025, July). Effect of continuous transition between force stimuli on haptic force discrimination tasks. In **2025 IEEE World Haptics Conference (WHC)** (pp. 71-77). doi: 10.1109/WHC64065.2025.11123225

Research Experience

- **Ph.D. Research — Haptic Force Perception and Interactive Systems** *Jan. 2022–Oct. 2026*
  IIT Jodhpur
  - Investigated human perception during interaction with **haptic and telepresence systems** through controlled psychophysical user studies.

- Designed and evaluated interactive experiments to study continuous force transitions, force-rate effects, multidimensional perceptual regions, perceived force intensity, and haptic delay perception.
- Quantitatively characterized human **perceptual thresholds** using psychometric functions, Weber fractions, JND estimation, machine-learning classification, and optimization techniques.
- Performed **statistical analysis** of human-subject data using normality tests, paired t-tests, ANOVA, and Friedman tests.
- Developed experience in designing interactive systems and evaluating them from the perspective of **human perception and behavior**.
- **Tools & Methods:** MATLAB, C++, Python, Visual Studio, CHAI3D, OpenHaptics SDK, PHANToM Premium 1.5 HF, Psychophysics, Classical Psychophysical Methods, Transformed Adaptive Methods, 2-AFC, 3I-3AFC, SVM, Simulated Annealing.

- **Research Projects** *Nov. 2022–Nov. 2024*
  IIT Jodhpur

  - **Telepresence and Teleaction System for Robot-Assisted Dentistry**
    * Designed and implemented a **virtual telepresence and teleaction (TPTA) system** for haptic interaction using the C++-based CHAI3D framework.
    * Designed and conducted a **psychophysical user study** to evaluate the **effects of round-trip delay and hand movement** on haptic perception.
    * Integrated haptic interaction, virtual environment modeling, and real-time force rendering for interactive teleoperation.
    * **Tools & Methods:** C++, Visual Studio, CHAI3D, PHANToM Premium 1.5 HF, MATLAB, Psychophysics, Adaptive Procedures.
  - **Haptics-Based Medical Simulator for Abdominal Palpation and Pulse Behavior**
    * Designed and implemented a **virtual environment and haptic interaction pipeline** for real-time force computation and rendering.
    * Implemented **haptic force-feedback and virtual viscoelastic object interaction** using the **Kelvin–Voigt Model** to simulate the force response of soft biological tissues during palpation.
    * **Tools & Methods:** C++, Visual Studio, CHAI3D, OpenHaptics SDK, PHANToM Premium 1.5 HF, Kelvin–Voigt Viscoelastic Model, Haptic Force-Feedback, Virtual Environment.

- **Mobile AR/VR Application Development and Virtual Simulation** *Jan. 2024–Apr. 2024; Jan. 2025–Apr. 2025*
  IIT Jodhpur

  - Conducted a hands-on **Mobile AR/VR development laboratory** covering Unity Game Engine concepts from fundamental to advanced levels, including scene development, asset integration, scripting, animation, interaction design, and real-time simulation.
  - Developed interactive **AR/VR environments using Unity and C#**, implementing 3D object interaction, physics-based behavior, user interfaces, and immersive virtual experiences.
  - Guided the development of Unity-based projects from basic interactive scenes to complete **AR/VR applications**, integrating 3D models, animations, interaction logic, and real-time rendering.
  - Developed, tested, and deployed final applications for **mobile and PC platforms**, providing hands-on experience in cross-platform application development and deployment.
  - **Tools & Methods:** Unity Game Engine, C#, AR/VR Development, 3D Modeling, Interactive Simulation, Physics-Based Interaction, UI/UX Design, Mobile and PC Application Development.

Technical Skills

- **Software & Programming:** MATLAB, Unity, C++, Python.
- **Haptic Tools & Libraries:** CHAI3D, OpenHaptics SDK, Phantom Premium 1.5 HF, Geomagic Touch.
- **Research Methods:** Psychophysics, Quantitative & Qualitative User Studies.
- **Data Analytics & Modeling:** Psychometric curve fitting, Statistical analysis, Machine-learning classification, Perceptual threshold estimation, Weber fraction analysis, and Psychophysical modeling.

Awards, Presentations & Activities

- **Oral Presentation** — *ACM Symposium on Applied Perception (ACM SAP 2026)* *Aug. 2026*
  Inria Centre at Rennes University, Rennes, France.
- **Oral Presentation** — *IEEE World Haptics Conference (IEEE WHC 2025)* *Jul. 2025*
  Suwon Convention Center, Suwon, Republic of Korea.
- **Poster Presentation** — *EuroHaptics 2026* *Jul. 2026*
  Previously Published Work Track, Siena, Italy.
- **Workshop Poster Presentation** — *EuroHaptics 2026* *Jul. 2026*
  Workshop: *From Research to Deployment: Standardization and Interoperability in Haptics*, Siena, Italy.
- **Poster Presentation** — *INAE-SERB Youth Conclave 2022* *Sep. 2022*
  Indian Institute of Technology Jodhpur, India.
- **International Travel Support (ITS)** — *Anusandhan National Research Foundation (ANRF)* *Jul. 2025*
  Awarded to attend *IEEE WHC 2025*, Suwon, Republic of Korea.
- **Student Volunteer** — *IEEE World Haptics Conference (WHC 2025)* *Jul. 2025*
  Suwon Convention Center, Suwon, Republic of Korea.
- **Student Volunteer** — *Next-gen AI: Inspiration from Brain Science (NAiBS 2023)* *Jan. 2023*
  Indian Institute of Technology Jodhpur, India.

Teaching Assistant

- Introduction to Haptics
- Psychophysics
- Mobile AR and VR
- Digital Signal Processing
- Signal & Systems
- Control System Lab

Academic Referees

Dr. Amit Bhardwaj *(Primary PhD Supervisor)*
Associate Professor
Department of Electrical Engineering
Indian Institute of Technology Jodhpur
Email: amitb@iitj.ac.in

Dr. Himanshu Kumar *(Co-Supervisor)*
Assistant Professor
Department of Electrical Engineering
Indian Institute of Technology Jodhpur
Email: hkumar@iitj.ac.in